Alexa Tinajero

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EDUCATION

University of North Carolina at Chapel Hill Expected May 2028

Bachelor's in Computer Science (Concentration: Mobile & Embedded Systems Engineering) | Minor in Artificial Intelligence

Extracurriculars: Study Abroad at Korea University, Engineering at Carolina (Vice President of R&D), LatinxEd Fellowship STEM Mentor, Omega Phi Alpha Service Sorority, Latinos in Tech / SHPE, NC Society of Hispanic Professionals (Lead STEM Ambassador)

Awards: Carolina Covenant Scholar, OUR Accelerated Research Award

EXPERIENCE

Experimental Engineering Laboratory & Interactive Robotics and Novel Technologies, UNC-Chapel Hill Dec 2025 – Present

Undergraduate Researcher & Control Systems Lead

- Engineer a six-stage voice-to-navigation pipeline for Boston Dynamics Spot to guide visually impaired users.
- Integrate speech-to-text, alias matching, and command validation with GraphNav autonomous navigation in Python.
- Map 50+ destinations across Sitterson Hall and automate map loading and localization for autonomous routing.
- Present the robotic seeing-eye assistant at 10+ events, including UNC's Celebration of Undergraduate Research.

Visual Computing and Augmented Intelligence Lab, UNC-Chapel Hill Sept 2024 – Feb 2025

Undergraduate Research Assistant

- Processed a 382 MB multi-camera imaging dataset, extracting frames from NetCDF (.nc) files with Python.
- Tested multi-camera image stitching in Hugin using frames extracted from the dataset.
- Reviewed computational imaging research and Python/Java code for neural-network-based feature detection.

UNC Lineberger Comprehensive Cancer Center, Dittmer Lab, UNC-Chapel Hill Aug 2023 – May 2024

Student Research Assistant

- Investigated drug response and resistance in Kaposi sarcoma and primary effusion lymphoma over 10 months.
- Wrote R and Python scripts for bioinformatics analysis of sequencing data from CRISPR-based experiments.
- Presented research findings at the NCSSM Annual Research Symposium to an audience of 500+ people.

PROJECTS

Plastic Recycling System Aug 2026 – Present

R&D Engineer & Project Manager

- Oversee a two-stage system that shreds discarded lab plastics and extrudes them into reusable 3D printing filament.
- Develop a compact shredder from Precious Plastic designs, selecting blades, shaft, motor, gearbox, and sieve size.
- Build a temperature-controlled, screw-fed filament extruder with diameter monitoring to produce printable 1.75 mm filament.

GutEase Aug 2026 – Present

Technical Consultant

- Advise a menstrual cramp relief startup on the technical design of a wearable with sensors and heating/cooling.
- Evaluate design modifications to reduce prototype size and weight, improving portability and comfort for the next iteration.
- Translate technical requirements into prototype specifications and milestones in preparation for the founder's investor pitch.

Voice-Controlled Spot Demo 2026

Robotics Software Developer

- Develop an offline voice-command pipeline in Python, integrating Vosk speech recognition with the Spot SDK.
- Implement command parsing and modular motion routines to translate spoken requests into robot actions.
- Integrate lidar person detection, obstacle checks, and pose tracking with AprilTag navigation fallbacks.

Personal Portfolio Website 2026

Software Developer & Designer

- Implemented section-based navigation and a rotating typewriter introduction highlighting my research and leadership roles.
- Separated site content from layout code using exported JavaScript objects and arrays to simplify updates.
- Illustrated original pixel art and incorporated it into the website's custom interface.

ADDITIONAL INFORMATION

Skills: Python, Java, C, C++, JavaScript, MATLAB, R, Boston Dynamics Spot SDK, GraphNav, Speech-to-Text

Tools: Git, GitHub, Docker, Linux/Unix, Node.js, VS Code, IntelliJ, Maven, GDB, RStudio, Excel

Certifications: Business Administration Certificate (WCC 2023), Lifeguard/CPR/AED/First Aid, PADI Open Water Scuba Diver

Interests: Community Service, Travel, Fashion, Photography, Ballet, Pilates, Boxing, Surfing, Marvel Rivals, PokémonGo